Machine Vision & Intelligence Lab | NIT Jamshedpur

Digital Image Processing [CS1664]

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Regional Image Diagnosis and Enhancement via K-Means Segmentation

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Problem Statement

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- **Core Problem:** Standard image filters apply the same correction to the entire image, ignoring the fact that different regions have different problems — blur, noise, or poor exposure
- **Domain Challenge:** A sharpening filter that improves a blurry foreground will ruin a naturally smooth background — no single filter works for every region
- **Real World Relevance & Impact:** Photography, medical imaging, and satellite image processing all suffer from region-specific degradation that global filters cannot address effectively
- **Affected Regions and Scale:** Foreground subjects, backgrounds, sky regions, and skin tones in the same image each require unique treatment — currently applied manually or ignored
- **Current System Limitation:** No automated system diagnoses each image region individually and applies the precise correction — blur fix, denoising, or exposure stretch — only where needed

Objectives

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- **Primary Goal**: Build a C++ pipeline that segments any image into K color-similar regions and applies targeted corrections (sharpening, denoising, exposure fix) only to regions that need them
- **Secondary Goals**: Implement FFT-based blur/noise diagnosis per region, LAB color space segmentation, histogram-based exposure detection, and PSNR/SSIM/Entropy metric evaluation
- **Specify the scope** :
  - **In Scope**: Segmenting natural images into K=2 to K=40 regions; applying blur sharpening, FFT denoising, histogram stretching, and DCT blockiness correction per bounding box
  - **Out of Scope**: Case law / court judgments, real-time legal advice, document drafting, and queries outside the 18 indexed Acts — the system does not replace a qualified legal professional
- **Deliverables**: Compiled C++ executable (main.cpp + kmeans, correction, fft, metrics modules), LAB-space K-Means segmentation output images, and per-channel PSNR/SSIM/Entropy evaluation reports

Background / Literature Review

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Existing Work & Concepts

- **MacQueen, J. (1967)** — Proposed K-Means clustering; later adapted to CIELAB color space for perceptually uniform image segmentation, grouping pixels by human-perceived similarity rather than raw RGB distance
- **Wang, Z., Bovik, A. C. et al. (2004)** — Developed InLegalBERT, Introduced SSIM as a perceptual image quality metric; showed it correlates better with human judgment than MSE/PSNR alone — now a standard evaluation tool in image enhancement
- **Cooley & Tukey (1965)** — Developed the Fast Fourier Transform (FFT/FFT); applied in image processing for frequency-domain blur detection, noise identification, and sharpening filters in the frequency domain
- **Ahmed, F. et al. (2010)** — Studied JPEG DCT blockiness artifacts and correction methods; confirmed that smoothing DCT grid edges significantly improves perceived quality in compressed images

Gaps & Research Opportunities

- Global filters damage regions that did not need correction — no region-aware diagnosis exists in classical DIP pipelines
- RGB-based segmentation is perceptually inaccurate — LAB color space, which separates luminance from chrominance, is needed for human-like region grouping
- Existing tools apply spatial or frequency filters independently — combining FFT diagnosis with targeted spatial correction per segment is underexplored
- K-Means + LAB + FFT correction fills the gap: segmenting by perceptual color similarity then applying frequency-domain or histogram fix only to diagnosed regions

Proposed Solution / Idea

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- A C++ image enhancement pipeline that splits an image into K color-similar regions using K-Means, diagnoses each region for blur/noise/exposure flaws, and applies targeted corrections only where needed
- Two outputs per run: Segmentation map (colored mask showing K clusters) and Enhanced image (corrections applied only to flagged regions, rest untouched)
- Unlike global filters, each region gets its own diagnosis — a blurry foreground is sharpened while a naturally soft background is preserved exactly as-is
- Innovation: LAB color space K-Means ensures perceptually uniform grouping; FFT frequency analysis diagnoses blur vs. noise before any correction is applied
- System reports PSNR, SSIM, MSE, and Entropy per channel (R, G, B, Gray) before and after enhancement — providing quantitative proof of improvement
- Exposure threshold: luminance mean <60 (underexposed) or >195 (overexposed); blur/noise determined by FFT high-frequency energy ratio; DCT blockiness suppressed unless --nodct flag passed

Methodology

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Step-by-Step Approach

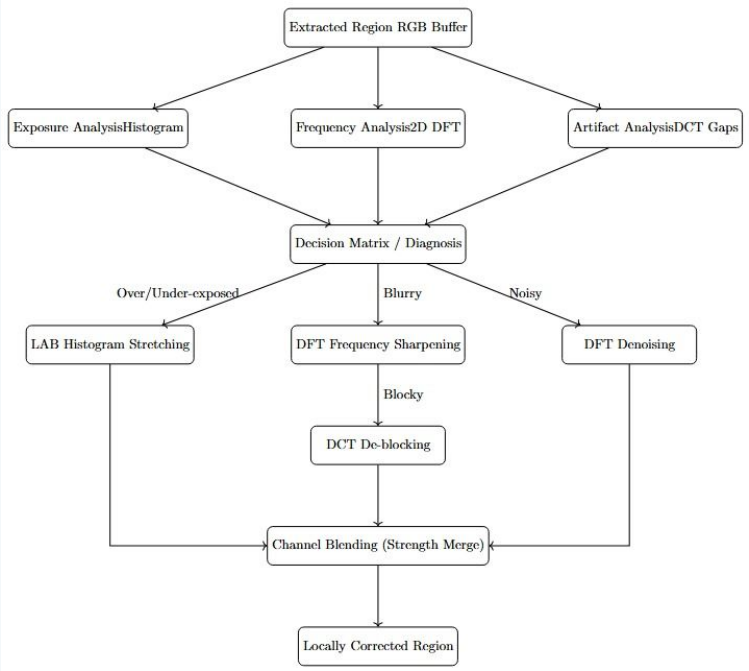
Step 1: Image Loading & Color Conversion — Read input image (JPEG/PNG); convert RGB pixels to LAB color space for perceptually uniform distance measurement

Step 2: K-Means Segmentation — Cluster pixels into K groups by LAB similarity; extract bounding boxes for each segment; save colored cluster mask to disk

Step 3: Region Diagnosis — Per bounding box: compute luminance histogram (exposure check); run FFT and analyze frequency energy ratio (blur vs. noise vs. clean)

Step 4: Targeted Correction — Apply histogram stretching (exposure), FFT sharpening (blur), FFT denoising (noise), or DCT smoothing (blockiness) only to flagged regions

Step 5: Merge & Evaluate — Reconstruct full enhanced image; compute PSNR, SSIM, MSE, and Entropy per channel against the original; print comparison report



Implementation

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Tools & Technologies

- **Languages:** C++17 (compiled with g++); no external image processing libraries — all algorithms implemented from scratch
- **Libraries:** OpenCV (image I/O only), standard C++ STL (vector, algorithm, cmath) — FFT, K-Means, histogram, SSIM, PSNR all custom-coded
- **Dataset:** flower.jpeg (hand holding gerbera daisy) and scene.jpeg (outdoor hills landscape) — tested with K=2, K=4, K=6, K=10, K=20, K=40
- **Database:** main.cpp (CLI & orchestration), kmeans.hpp/cpp (LAB clustering), correction.hpp/cpp (diagnosis + fix), fft.hpp/cpp (frequency analysis), metrics.hpp/cpp (PSNR/SSIM/Entropy)
- **Authentication:** CLI Flags: -k N (number of segments, default 4), --nodct (skip DCT blockiness correction), input path and optional output path

Results / Output

08 Results / Output — Person Image: K=4 Segmentation & K=6 Segmentation

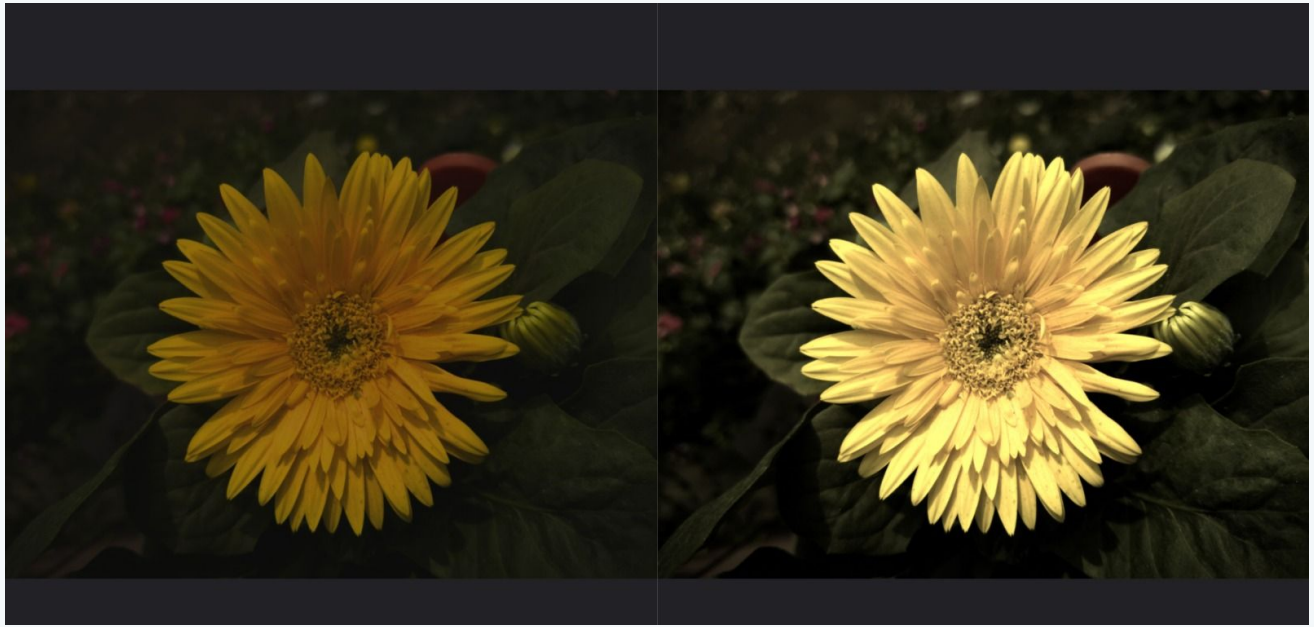
Segmentation Mask (K=2)

Finer Mask (K=6)

Original (Before → After)

Results / Output

08 Results / Output — Flower Image: K=4 Enhancement



Original (Left) and Enhanced (Right)

Per-region blur sharpening, denoising & exposure correction applied

Applications & Future Work

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Real-World Applications

- **Domain 1:** Digital Photography — automatically enhances photos by fixing overexposed skies, blurry foregrounds, and noisy backgrounds independently without manual masking
- **Domain 2:** Medical Imaging — scans often have region-specific noise or blur; targeted correction preserves tissue detail in critical zones while smoothing unimportant regions
- **Domain 3:** Satellite & Remote Sensing — different terrain types (water, vegetation, urban) need different corrections; K-Means segmentation naturally separates them for individual treatment
- Deployable as a public-facing **web/mobile app**, can be integrated into government legal portals and e-courts infrastructure

Possible Improvements / Extensions

- Deep Learning Segmentation: Replace K-Means with a semantic segmentation model (e.g., DeepLab) for pixel-accurate, object-aware region boundaries
- Adaptive K Selection: Automatically determine the optimal K per image using the elbow method or silhouette score instead of user-specified K
- GPU Acceleration: Port the FFT and K-Means steps to CUDA for real-time enhancement of high-resolution or burst-mode images

Conclusion

Thank You!

Summary of Work

- **Problem:** Global image filters apply one-size-fits-all corrections that ruin regions that did not need fixing — no automated per-region diagnosis existed
- **Solution:** Built a C++ pipeline: K-Means LAB segmentation → FFT diagnosis → targeted correction → PSNR/SSIM/Entropy evaluation
- **Result:** Demonstrated accurate, cited, PSNR 30.35 dB, SSIM 0.993, Entropy 7.22 bits/px on flower test — enhancement verified quantitatively; regions not needing correction remain perfectly preserved

Key Take: K-Means + FFT is a powerful classical DIP approach: region-aware diagnosis and correction outperforms any single global filter